1 Gain law

(bars denote normalized quantities: lengths by R; $a_o := \Delta t/(\gamma_N R)$ [1/pN] and $a_{\text{nom}} := \Delta t/\gamma_N = a_o R$ [μm/pN]; $\bar{a} := a/a_{\text{nom}}$; forces by a_o: $\bar{f} := a_o f$, so that $\bar{a}\,\bar{f} = \Delta\bar{w}$; $0 \le \bar{a} \le 1$, far field 1, contact 0; hats sit on the normalized quantities $\hat{\bar{a}}_w$, $\hat{b}$, $\delta\hat{\bar{w}}$; $e_\theta = \theta - \hat{\theta}$)

$$
\begin{aligned}
\bar{a}(\bar{w}) &= 1 - \frac{1}{b\,(\bar{w} - \bar{w}_s)}\\
\bar{a}'(\bar{a}, b) &:= \frac{d\bar{a}}{d\bar{w}} = b\big(1 - \bar{a}\big)^2\\
\bar{a}''(\bar{a}, b) &:= \frac{d\bar{a}'}{d\bar{w}} = -2\,b^2\big(1 - \bar{a}\big)^3\\
a_w &= a_o\,\bar{a}_w, \qquad a_o = \frac{\Delta t}{\gamma_N R}
\end{aligned}
$$

2 True dynamics

$\big(\delta\bar{w}_j[k] := \delta\bar{w}[k - d + j - 1],\ j = 1, \ldots, d+1,\ d = 2\big)$

$$
\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k]\\
\bar{w}_d[k+1] &= \bar{w}_d[k] + \Delta\bar{w}_d[k]\\
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k]\\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k]\\
\delta\bar{w}_3[k+1] &= \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k]\\
\Delta\bar{w}[k] &= \Delta\bar{w}_d[k] + (1 - \lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k]\\
\bar{a}_w[k+1] &= \bar{a}_w[k] + b[k]\big(1 - \bar{a}_w[k]\big)^2\,\Delta\bar{w}[k]\\
b[k+1] &= b[k]
\end{aligned}
$$

$$
\begin{aligned}
F_{dw}[k] &= \bar{f}_{dw}[k] + (1 - \lambda_c)\sum_{i=1}^{d}\bar{f}_{dw}[k-i]\\
\varepsilon_w[k] &= \bar{a}_w[k]\,\bar{f}_T[k] + (1 - \lambda_c)\sum_{i=1}^{d}\bar{a}_w[k-i]\,\bar{f}_T[k-i] - (1 - \lambda_c)\,n_w[k]
\end{aligned}
$$

3 Estimator

$$
\mathbf{x} = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ b\,\big]^\top \qquad (d = 2)
$$

$$
\begin{aligned}
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\, \delta\hat{\bar{w}}_3[k] \\
\Delta\hat{\bar{w}}[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\, \delta\hat{\bar{w}}_3[k] \\
\hat{\bar{a}}_w[k+1] &= \hat{\bar{a}}_w[k] + \hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)^2\, \Delta\hat{\bar{w}}[k] \\
\hat{b}[k+1] &= \hat{b}[k]
\end{aligned}
$$

4 Error dynamics

(True − Estimator; coefficients at the estimate)

$$
\begin{aligned}
e_{\Delta\bar{w}}[k] = \Delta\bar{w}[k] - \Delta\hat{\bar{w}}[k] &= (1-\lambda_c)\, e_{\delta\bar{w}_3}[k] + F_{dw}[k]\, e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
b[k]\big(1-\bar{a}_w[k]\big)^2 &= \hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)^2 - 2\,\hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)\, e_{\bar{a}_w}[k] + \big(1-\hat{\bar{a}}_w[k]\big)^2 e_b[k] + \mathcal{O}\big(e^2\big)
\end{aligned}
$$

$$
\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c\, e_{\delta\bar{w}_3}[k] - F_{dw}[k]\, e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= \Big[1 + \hat{b}\big(1-\hat{\bar{a}}_w\big)^2 F_{dw} - 2\hat{b}\big(1-\hat{\bar{a}}_w\big)\big(\Delta\bar{w}_d + (1-\lambda_c)\, \delta\hat{\bar{w}}_3\big)\Big] e_{\bar{a}_w}[k] \\
&\quad + (1-\lambda_c)\, \hat{b}\big(1-\hat{\bar{a}}_w\big)^2 e_{\delta\bar{w}_3}[k] + \big(1-\hat{\bar{a}}_w\big)^2 \big(\Delta\bar{w}_d + (1-\lambda_c)\, \delta\hat{\bar{w}}_3\big) e_b[k] + \hat{b}\big(1-\hat{\bar{a}}_w\big)^2 \varepsilon_w[k] \\
e_b[k+1] &= e_b[k]
\end{aligned}
$$

$$
F_e[k] = \begin{bmatrix}
0 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 \\
0 & 0 & \lambda_c & -F_{dw} & 0 \\
0 & 0 & (1-\lambda_c)\, \hat{b}\big(1-\hat{\bar{a}}_w\big)^2 & \begin{array}{l} 1 + \hat{b}\big(1-\hat{\bar{a}}_w\big)^2 F_{dw} \\ \quad -2\hat{b}\big(1-\hat{\bar{a}}_w\big)\big(\Delta\bar{w}_d + (1-\lambda_c)\, \delta\hat{\bar{w}}_3\big) \end{array} & \begin{array}{l} \big(1-\hat{\bar{a}}_w\big)^2 \\ \quad \times\big(\Delta\bar{w}_d + (1-\lambda_c)\, \delta\hat{\bar{w}}_3\big) \end{array} \\
0 & 0 & 0 & 0 & 1
\end{bmatrix}
$$

$$
\mathbf{q}[k] = \Big[0,\ 0,\ -\varepsilon_w,\ \hat{b}\big(1-\hat{\bar{a}}_w\big)^2 \varepsilon_w,\ 0\Big]^\top
$$